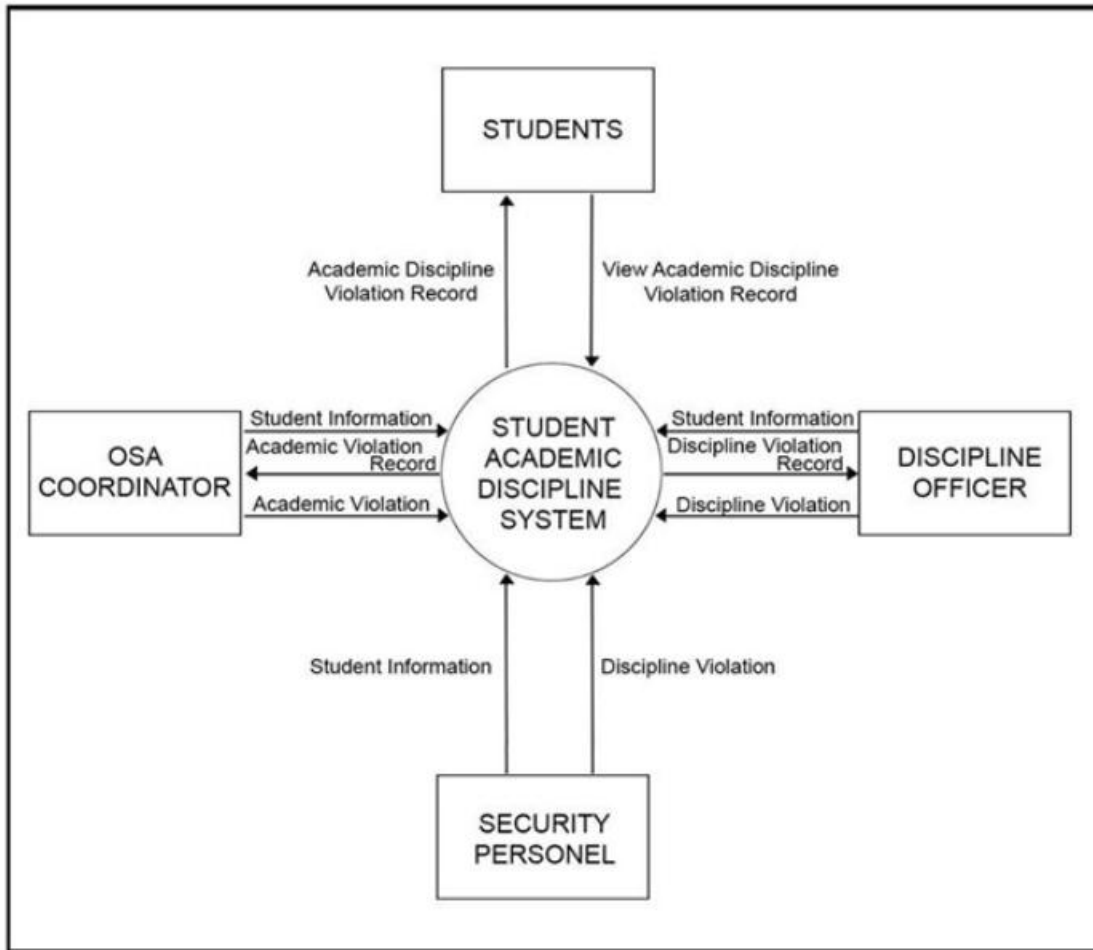


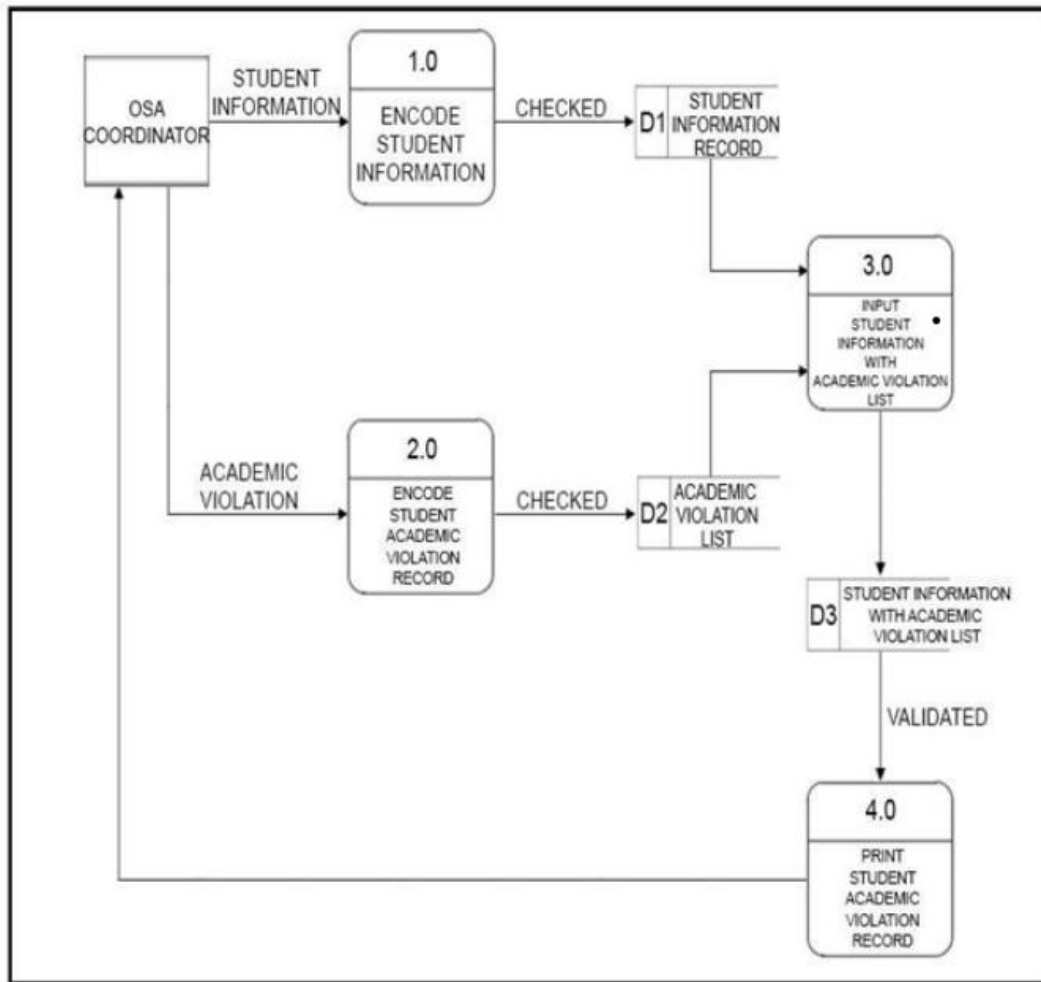
reference – osa Management System

link = <https://itsourcecode.com/uml/food-ordering-system-dfd/>

DFD level 0



Level 1



Todo-Vanguard ATM System

Opo mahal. Let's finalize ****our original DFD****, with only the formatting and data-flow simplification you requested.

Enterprise-Kiosk Vanguard

Final Data Flow Diagram

Shape / Symbol Legend

Label	Meaning
-----	-----
`*ENTITY*`	External Entity
`*PROCESS*`	Process / action
`*DATA FLOW*`	Data in motion
`*DATA STORE*`	Stored data

LEVEL 0 — CONTEXT DIAGRAM

****System Process:**** `PROVIDE ATM SERVICES`

External Entities

* `\*ENTITY\*` CUSTOMER`

* `\*ENTITY\*` CORE BANKING SYSTEM`

* `\*ENTITY\*` FLEET OPERATIONS CENTER`

Final Level 0

```text

```

 ENTITY
CORE BANKING SYSTEM
 |
 | *DATA FLOW*
```

| Account Data

| Authorization



**\*PROCESS\***

**PROVIDE ATM SERVICES**

|

| **\*DATA FLOW\***

| Transaction Data

| Account Updates



**\*ENTITY\***

**CORE BANKING SYSTEM**

**\*ENTITY\* CUSTOMER**

|

| **\*DATA FLOW\***

| Biometric Data

| Gesture Input

| Transaction Request

| Security Request



**\*PROCESS\***

**PROVIDE ATM SERVICES**

|

| **\*DATA FLOW\***

| Authentication

- | Dashboard
- | Transaction Result
- | Cash Status
- | Receipt



**\*ENTITY\* CUSTOMER**

**\*PROCESS\***

**PROVIDE ATM SERVICES**

- |
- | **\*DATA FLOW\***
- | Telemetry
- | Cash Status
- | Maintenance Status



**\*ENTITY\***

**FLEET OPERATIONS CENTER**

- |
- | **\*DATA FLOW\***
- | Dispatch Status
- | Maintenance Status



**\*PROCESS\***

**PROVIDE ATM SERVICES**

...

### ### Level 0 Data Flows — Final Prepared Terms

| Direction              | Data Flow               |
|------------------------|-------------------------|
| -----                  | -----                   |
| Customer → ATM         | **Biometric Data**      |
| Customer → ATM         | **Gesture Input**       |
| Customer → ATM         | **Transaction Request** |
| Customer → ATM         | **Security Request**    |
| ATM → Customer         | **Authentication**      |
| ATM → Customer         | **Dashboard**           |
| ATM → Customer         | **Transaction Result**  |
| ATM → Customer         | **Cash Status**         |
| ATM → Customer         | **Receipt**             |
| Core Banking → ATM     | **Account Data**        |
| Core Banking → ATM     | **Authorization**       |
| ATM → Core Banking     | **Transaction Data**    |
| ATM → Core Banking     | **Account Updates**     |
| ATM → Fleet Operations | **Telemetry**           |
| ATM → Fleet Operations | **Cash Status**         |
| ATM → Fleet Operations | **Maintenance Status**  |
| Fleet Operations → ATM | **Dispatch Status**     |
| Fleet Operations → ATM | **Maintenance Status**  |

---

### # LEVEL 1 — DETAILED DFD

The original four processes are retained:

| Process        | Action                           |
|----------------|----------------------------------|
| -----          | -----                            |
| <b>**1.0**</b> | <b>**Authenticate User**</b>     |
| <b>**2.0**</b> | <b>**Process Transaction**</b>   |
| <b>**3.0**</b> | <b>**Monitor ATM Resources**</b> |
| <b>**4.0**</b> | <b>**Protect ATM Session**</b>   |

### ### Data Stores

| Data Store    | Name                              |
|---------------|-----------------------------------|
| -----         | -----                             |
| <b>**D1**</b> | <b>Customer / Biometric Data</b>  |
| <b>**D2**</b> | <b>Account / Transaction Data</b> |
| <b>**D3**</b> | <b>ATM Terminal Data</b>          |
| <b>**D4**</b> | <b>Cassette Inventory</b>         |
| <b>**D5**</b> | <b>Maintenance Records</b>        |

---

### ## 1.0 Authenticate User

```text

ENTITY

CUSTOMER

|

| ***DATA FLOW***

| **Biometric Data**



PROCESS

1.0 AUTHENTICATE USER



| ***DATA FLOW***

| **Biometric Profile**



DATA STORE

D1 CUSTOMER / BIOMETRIC DATA

PROCESS

1.0 AUTHENTICATE USER



| ***DATA FLOW***

| **Authentication**



ENTITY

CUSTOMER

PROCESS

1.0 AUTHENTICATE USER



| ***DATA FLOW***

| **User Context**



PROCESS

2.0 PROCESS TRANSACTION

...

2.0 Process Transaction

```text

**\*ENTITY\***

**CUSTOMER**

|

| **\*DATA FLOW\***

| **Gesture Input**

| **Transaction Request**

▼

**\*PROCESS\***

**2.0 PROCESS TRANSACTION**

|

| **\*DATA FLOW\***

| **Account Data**

▼

**\*DATA STORE\***

**D2 ACCOUNT / TRANSACTION DATA**

**\*ENTITY\***

## CORE BANKING SYSTEM

|

| **\*DATA FLOW\***

| Account Data

| Authorization



**\*PROCESS\***

## 2.0 PROCESS TRANSACTION

**\*PROCESS\***

## 2.0 PROCESS TRANSACTION

|

| **\*DATA FLOW\***

| Transaction Data

| Account Updates



**\*ENTITY\***

## CORE BANKING SYSTEM

**\*PROCESS\***

## 2.0 PROCESS TRANSACTION

|

| **\*DATA FLOW\***

| Dashboard

| Transaction Result

| Cash Status

| Receipt



**\*ENTITY\***

**CUSTOMER**

...

---

### # 3.0 Monitor ATM Resources

```text

DATA STORE

D3 ATM TERMINAL DATA

|

| ***DATA FLOW***

| Terminal Data



PROCESS

3.0 MONITOR ATM RESOURCES

DATA STORE

D4 CASSETTE INVENTORY

|

| ***DATA FLOW***

| Cash Data



PROCESS

3.0 MONITOR ATM RESOURCES

DATA STORE

D5 MAINTENANCE RECORDS

|

| ***DATA FLOW***

| **Maintenance Data**



PROCESS

3.0 MONITOR ATM RESOURCES

PROCESS

3.0 MONITOR ATM RESOURCES

|

| ***DATA FLOW***

| **Telemetry**

| **Cash Status**

| **Maintenance Status**



ENTITY

FLEET OPERATIONS CENTER

ENTITY

FLEET OPERATIONS CENTER

|

| *DATA FLOW*

| Dispatch Status

| Maintenance Status



PROCESS

3.0 MONITOR ATM RESOURCES

...

4.0 Protect ATM Session

```text

**\*ENTITY\***

**CUSTOMER**

|

| \*DATA FLOW\*

| Security Request



**\*PROCESS\***

### 4.0 PROTECT ATM SESSION

|

| \*DATA FLOW\*

| Security Status



**\*ENTITY\***

**CUSTOMER**

**\*PROCESS\***

**4.0 PROTECT ATM SESSION**

|

| **\*DATA FLOW\***

| Security Data



**\*DATA STORE\***

**D3 ATM TERMINAL DATA**

...

---

**# FINAL LEVEL 1 — DATA FLOW SUMMARY**

This is the **\*\*clean list to use when drawing the arrows\*\***:

| From                  | Data Flow                    | To                         |  |
|-----------------------|------------------------------|----------------------------|--|
| -----                 | -----                        | -----                      |  |
| Customer              | <b>**Biometric Data**</b>    | 1.0 Authenticate User      |  |
| 1.0 Authenticate User | <b>**Biometric Profile**</b> | D1 Customer/Biometric Data |  |
| 1.0 Authenticate User | <b>**Authentication**</b>    | Customer                   |  |
| 1.0 Authenticate User | <b>**User Context**</b>      | 2.0 Process Transaction    |  |
| Customer              | <b>**Gesture Input**</b>     | 2.0 Process Transaction    |  |

|                             |                                |                           |  |
|-----------------------------|--------------------------------|---------------------------|--|
| Customer                    | <b>**Transaction Request**</b> | 2.0 Process Transaction   |  |
| D2 Account/Transaction Data | <b>**Account Data**</b>        | 2.0 Process Transaction   |  |
| 2.0 Process Transaction     | <b>**Authorization**</b>       | Core Banking System       |  |
| Core Banking System         | <b>**Account Data**</b>        | 2.0 Process Transaction   |  |
| 2.0 Process Transaction     | <b>**Transaction Data**</b>    | Core Banking System       |  |
| 2.0 Process Transaction     | <b>**Account Updates**</b>     | Core Banking System       |  |
| 2.0 Process Transaction     | <b>**Dashboard**</b>           | Customer                  |  |
| 2.0 Process Transaction     | <b>**Transaction Result**</b>  | Customer                  |  |
| 2.0 Process Transaction     | <b>**Cash Status**</b>         | Customer                  |  |
| 2.0 Process Transaction     | <b>**Receipt**</b>             | Customer                  |  |
| D3 ATM Terminal Data        | <b>**Terminal Data**</b>       | 3.0 Monitor ATM Resources |  |
| D4 Cassette Inventory       | <b>**Cash Data**</b>           | 3.0 Monitor ATM Resources |  |
| D5 Maintenance Records      | <b>**Maintenance Data**</b>    | 3.0 Monitor ATM Resources |  |
| 3.0 Monitor ATM Resources   | <b>**Telemetry**</b>           | Fleet Operations Center   |  |
| 3.0 Monitor ATM Resources   | <b>**Cash Status**</b>         | Fleet Operations Center   |  |
| 3.0 Monitor ATM Resources   | <b>**Maintenance Status**</b>  | Fleet Operations Center   |  |
| Fleet Operations Center     | <b>**Dispatch Status**</b>     | 3.0 Monitor ATM Resources |  |
| Fleet Operations Center     | <b>**Maintenance Status**</b>  | 3.0 Monitor ATM Resources |  |
| Customer                    | <b>**Security Request**</b>    | 4.0 Protect ATM Session   |  |
| 4.0 Protect ATM Session     | <b>**Security Status**</b>     | Customer                  |  |
| 4.0 Protect ATM Session     | <b>**Security Data**</b>       | D3 ATM Terminal Data      |  |

**### Final process names**

**\*\*1.0 Authenticate User\*\***

**\*\*2.0 Process Transaction\*\***

**\*\*3.0 Monitor ATM Resources\*\***

**\*\*4.0 Protect ATM Session\*\***

**### Final prepared data-flow vocabulary**

**\*\*Biometric Data • Biometric Profile • Authentication • User Context • Gesture Input • Transaction Request • Account Data • Authorization • Transaction Data • Account Updates • Dashboard • Transaction Result • Cash Status • Receipt • Terminal Data • Cash Data • Maintenance Data • Telemetry • Maintenance Status • Dispatch Status • Security Request • Security Status • Security Data\*\***

This is now the **\*\*finalized DFD vocabulary and structure\*\*** for our Enterprise-Kiosk Vanguard case, while retaining the original DFD design.

---

## **ERD Ms Access**

---

Opo mahal. We'll make this specifically for **\*\*Microsoft Access SQL\*\***, using your naming convention and all **\*\*9 tables\*\***, with the **\*\*PK/FK relationships\*\*** included.

One important Access detail: because some Access versions can be picky about executing multiple ``CREATE TABLE`` statements at once, **\*\*run each table statement separately first\*\***, then run each ``ALTER TABLE`` relationship statement. That avoids the naming/parser errors you're encountering.

## **## 1. Create `tblCustomers`**

Open **\*\*Create → Query Design → SQL View\*\***, paste this, then click **\*\*Run\*\***:

**```sql**

```
CREATE TABLE tblCustomers (
 numCustomerID AUTOINCREMENT CONSTRAINT PK_tblCustomers PRIMARY
KEY,
 txtFirstName TEXT(50),
 txtLastName TEXT(50),
 txtContactNumber TEXT(20),
 txtEmail TEXT(100),
 dteRegistrationDate DATETIME,
 txtStatus TEXT(20)
);
```
```

2. Create `tblBiometricTypes`

```sql

```
CREATE TABLE tblBiometricTypes (  
    numBiometricTypeID AUTOINCREMENT CONSTRAINT PK_tblBiometricTypes  
PRIMARY KEY,  
    txtBiometricTypeName TEXT(50),  
    txtDescription TEXT(255)  
);
```

3. Create `tblBiometricProfiles`

```sql

```
CREATE TABLE tblBiometricProfiles (
 numBiometricProfileID AUTOINCREMENT CONSTRAINT
 PK_tblBiometricProfiles PRIMARY KEY,
 numCustomerID LONG,
 numBiometricTypeID LONG,
 memBiometricTemplate LONGTEXT,
 dteEnrollmentDate DATETIME,
 txtStatus TEXT(20)
);
```

---

---

### **## 4. Create `tblAccounts`**

```sql

```
CREATE TABLE tblAccounts (  
    numAccountID AUTOINCREMENT CONSTRAINT PK_tblAccounts PRIMARY  
    KEY,  
    numCustomerID LONG,  
    txtAccountNumber TEXT(30),
```

```
txtAccountType TEXT(30),  
curCurrentBalance CURRENCY,  
txtAccountStatus TEXT(20),  
dteDateOpened DATETIME  
);  
---
```

5. Create `tblTransactionTypes`

```
```sql  
CREATE TABLE tblTransactionTypes (
 numTransactionTypeID AUTOINCREMENT CONSTRAINT
 PK_tblTransactionTypes PRIMARY KEY,
 txtTransactionTypeName TEXT(50),
 txtDescription TEXT(255)
);

```

**## 6. Create `tblATMTerminals`**

```
```sql  
CREATE TABLE tblATMTerminals (  
    numATMID AUTOINCREMENT CONSTRAINT PK_tblATMTerminals PRIMARY  
    KEY,
```

```
txtTerminalSerialNumber TEXT(50),  
txtLocation TEXT(150),  
txtStatus TEXT(20),  
dteInstallationDate DATETIME  
);  
---
```

7. Create `tblATMTransactions`

```
```sql  
CREATE TABLE tblATMTransactions (
 numTransactionID AUTOINCREMENT CONSTRAINT PK_tblATMTransactions
 PRIMARY KEY,
 numAccountID LONG,
 numATMID LONG,
 numTransactionTypeID LONG,
 curAmount CURRENCY,
 dteTransactionDateTime DATETIME,
 txtTransactionStatus TEXT(20),
 txtReceiptType TEXT(30)
);

```

## **## 8. Create `tblCassetteInventory`**

**```sql**

```
CREATE TABLE tblCassetteInventory (
 numCassetteID AUTOINCREMENT CONSTRAINT PK_tblCassetteInventory
PRIMARY KEY,
 numATMID LONG,
 curDenomination CURRENCY,
 numCurrentQuantity LONG,
 numMaximumCapacity LONG,
 txtCassetteStatus TEXT(20),
 dteLastRefilledDate DATETIME
);
```

**```**

**---**

## **## 9. Create `tblMaintenanceRecords`**

**```sql**

```
CREATE TABLE tblMaintenanceRecords (
 numMaintenanceID AUTOINCREMENT CONSTRAINT
PK_tblMaintenanceRecords PRIMARY KEY,
 numATMID LONG,
 txtMaintenanceType TEXT(50),
 dteMaintenanceDate DATETIME,
 memDescription LONGTEXT,
 txtTechnicianName TEXT(100),
```

```
txtMaintenanceStatus TEXT(20)
);
...
```

**# Now create the PK/FK relationships**

**After all \*\*9 tables have been successfully created\*\*, create the relationships.**

**### Relationship 1 — Customer → Biometric Profiles**

```
```sql
ALTER TABLE tblBiometricProfiles
ADD CONSTRAINT FK_BiometricProfiles_Customers
FOREIGN KEY (numCustomerID)
REFERENCES tblCustomers (numCustomerID);
...
```

Relationship 2 — Biometric Type → Biometric Profiles

```
```sql
ALTER TABLE tblBiometricProfiles
ADD CONSTRAINT FK_BiometricProfiles_BiometricTypes
FOREIGN KEY (numBiometricTypeID)
REFERENCES tblBiometricTypes (numBiometricTypeID);
...
```

**### Relationship 3 — Customer → Accounts**

```
```sql
```

```
ALTER TABLE tblAccounts  
ADD CONSTRAINT FK_Accounts_Customers  
FOREIGN KEY (numCustomerID)  
REFERENCES tblCustomers (numCustomerID);
```

```
```
```

**### Relationship 4 — Account → ATM Transactions**

```
```sql
```

```
ALTER TABLE tblATMTransactions  
ADD CONSTRAINT FK_ATMTransactions_Accounts  
FOREIGN KEY (numAccountID)  
REFERENCES tblAccounts (numAccountID);
```

```
```
```

**### Relationship 5 — Transaction Type → ATM Transactions**

```
```sql
```

```
ALTER TABLE tblATMTransactions  
ADD CONSTRAINT FK_ATMTransactions_TransactionTypes  
FOREIGN KEY (numTransactionTypeID)  
REFERENCES tblTransactionTypes (numTransactionTypeID);
```

```
```
```

**### Relationship 6 — ATM Terminal → ATM Transactions**

```
```sql
```

```
ALTER TABLE tblATMTransactions  
ADD CONSTRAINT FK_ATMTransactions_ATMTerminals  
FOREIGN KEY (numATMID)  
REFERENCES tblATMTerminals (numATMID);
```

```
```
```

**### Relationship 7 — ATM Terminal → Cassette Inventory**

```
```sql
```

```
ALTER TABLE tblCassetteInventory  
ADD CONSTRAINT FK_CassetteInventory_ATMTerminals  
FOREIGN KEY (numATMID)  
REFERENCES tblATMTerminals (numATMID);
```

```
```
```

**### Relationship 8 — ATM Terminal → Maintenance Records**

```
```sql
```

```
ALTER TABLE tblMaintenanceRecords  
ADD CONSTRAINT FK_MaintenanceRecords_ATMTerminals  
FOREIGN KEY (numATMID)  
REFERENCES tblATMTerminals (numATMID);
```

```
```
```

**## Final relationship structure**

```
```text
```

tblCustomers

```
```
```

**### One thing to be careful about, mahal**

**For these foreign-key fields:**

```
```text
```

numCustomerID

numBiometricTypeID

numAccountID

numATMID

numTransactionTypeID

```
```
```

**the \*\*data type is `LONG` (Number)\*\*, not AutoNumber.**

**Only the corresponding primary keys are `AUTOINCREMENT`:**

```
```text
```

numCustomerID → AutoNumber

numBiometricTypeID → AutoNumber

numBiometricProfileID → AutoNumber

numAccountID → AutoNumber

numTransactionTypeID → AutoNumber

numATMID → AutoNumber

numTransactionID → AutoNumber

numCassetteID → AutoNumber

numMaintenanceID → AutoNumber

...

That distinction is important for Access relationships.

****Also, don't execute all 17 SQL blocks together.** Create the 9 tables first, one query at a time, and only after all tables exist, execute the 8 relationship queries. This is the safest way to avoid another Access SQL error.**

Search Key words:

2. Complete ERD with Entities, PK/FKs, Attributes, & Cardinality Notations

- at least these 6 entities

3: Low-Fidelity (Low-Fi) HCI Physical & Software Prototype

-